

20240909 EB_MN (solubility)

Project: Yi-Shuan Tseng Holt Lab
Author: Yi Shuan Tseng
Status: In progress
Entry Created On: 25 Sep 2024 04:09:40 UTC
Entry Last Modified: 06 May 2025 21:03:09 UTC
Export Generated On: 01 Sep 2026 18:30:27 UTC

- 2024年9月9日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2024年9月11日 星期三
- T2 -> T3
- 2024年9月13日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2024年9月16日 星期一
- T4 -> T5
- 2024年9月18日 星期三
- T5 -> T6
- 2024年9月20日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2024年9月23日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 2. Remove the PLO and wash the plate with PBS once
 3. Coat the plate with laminine for 1hr.
 4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
 6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
 7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
 8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. 1.08 M/6-=well

Well1

	1	2	3
A	BP	hypo 1h	hypo 24h
B	BP+24h rescue	hypo1h+24h rescue	hypo24h+24h rescue

2024年9月25日 星期三

T7-> BP T8

2024年9月27日 星期五

400uL BP T8 -> 500 uL BP T8

2024年9月30日 星期一

400uL BP T8 -> 500 uL BP T8

2024年10月1日 星期二

treat hypo for B3 at 2pm

2024年10月2日 星期三

400uL BP T8 -> 500 uL BP T8 for A1-3 (made a mistake here. I should change A3 to hypo)

B1: BP at 12:50 pm, BP at 1:50 pm

B2: hypo (0.15mM) at 12:50 pm, BP at 1:50 pm

2024年10月3日 星期四

A1: 1:15 pm BP

A2 & A3: 1:15 pm hypo

updated well plan

	1	2	3
A	BP	hypo 1h	hypo 1h
B	BP+24h rescue	hypo1h+24h rescue	hypo24h+24h rescue

2024年10月4日 星期五

conduct soluble and insoluble protein extraction: see protocol

use 50 uL Urea+DTT to insoluble pellet, but the pellet is really hard to break down

store protein in -80C

Soluble and insoluble protein extraction and blotting

Introduction

This is the protocol for soluble and insoluble fractionation, from Francesco Limone in his times in the Eggan lab. Ref: [Klim et al., Nat Neurosci, 2019](#)

Materials

- › **RIPA Buffer** prepared on Tuesday, as follows: RIPA (150 mM NaCl, 1.0% IGEPAL® CA-630, 0.5% sodium deoxycholate, 0.1% SDS, 50 mM Tris, pH 8.0). 2 ml aliquotes prepared and stocked into box "RIPA/Urea buffers" in -80 Rack 2 shelf 2 (from the top).
- › **Urea Buffer** prepared using CSH Protocol, as follows: 9M Urea, 4% CHAPS, 50 mM DTT (not added) in water. 2 ml aliquotes prepared and stocked into box "RIPA/Urea buffers" in -80 Rack 2 shelf 2 (from the top).
- › Cell scraper
- › EtOH 70%
- › Bioruptor Diogenade
- › Chilled centrifuge (4C)
- › SurePAGE, Bis-Tris, 4-12% (Genscript #M00653)
- › MES SDS Running Buffer Powder (Genscript #M00677)
- › 10x Transfer Buffer
- › Methanol
- › 5% BSA in TBS-T
- › Powder milk

›

Procedure

Protocol Cell Lysis

- ✓ 1. Wash cells once with cold PBS
- ✓ 2. Remove PBS and add 150 µL of **RIPA buffer with protease inhibitors** to each well of a 6-well plate

- ✓ 3. Incubate 15 minutes on ice with agitation
- ✓ 4. Scrape each well, tilt plate in ice and use the scraper to dislodge the cells to collect at the bottom of the plate.
- ✓ 5. Transfer lysate to microcentrifuge tube.
- ✓ 6. Sonicate lysates in **Bioruptor Diagenade mode M, 20 sec on, 20 sec off, 20 cycles (15 min)**
- ✓ 7. Spin lysates in 4°C centrifuge for 10 min at 17,000 x g.
- ✓ 8. Transfer supernatant to a new tube. The supernatant is the soluble lysate and the pellet is the insoluble lysate.
- ✓ 9.  **CRITICAL** The pellet is really small, always make sure that you see it before removing the supernatant.
- ✓ 10. Run BCA on soluble lysate to determine its concentration.
- ✓ 11. For the insoluble component, resuspend in 250 uL of RIPA buffer and Sonicate in Bioruptor Diagenade mode M, 20 sec on, 20 sec off, 20 cycles (15 min)
- ✓ 12. Spin lysates in 4°C centrifuge for 7 min at 17,000 x g.
- ✓ 13. Perform wash 1 more time.
- ✓ 14. Spin lysates in 4°C centrifuge for 7 min at 17,000 x g.
- ✓ 15. Then resuspend in 50ul of **Urea Buffer and Sonicate** in Bioruptor Diagenade mode M, 20 sec on, 20 sec off, 20 cycles (15 min).
- ✓ 16.  **CRITICAL** The pellet might not dissolve completely, but that is ok. Just make sure to get the pellet undissolved when you prepare the blue.
- ✓ 17.  **PAUSE** Store the samples at -80 until use.

SDS Electrophoresis and Western Blot

- ✓ 18. Prepare the blues to have 10/20ug of protein in total, according to the concentration measured in the soluble fraction.
- ✓ 19.  **PAUSE** You can store the blues at -20 for up to 2 weeks before performing the WB.
- ✓ 20.  **CRITICAL** Warm the blues: 15 min at 95C for the Soluble fraction; 15 min at 37 for the Insoluble fractions (it crosslinks if done at more than 37)
- ✓ 21. Load a SurePAGE, Bis-Tris, 4-12% (Genscript #M00653) with 5/10 ug of protein each well.

- ✓ 22. Run in MES SDS Running Buffer for around 1.5 hours, at 120 mV
- ✓ 23. Preactivated 0.45 um PVDF membrane for 30 seconds in pure methanol, then rehydrate it in milliQ water, and keep it in 1x transfer buffer until use.
- ✓ 24. Transfer for 1h at 100 mV constant in the cold room. Prepare 1x transfer buffer with the 10x Transfer buffer and 20% methanol.
- ✓ 25. Block membrane for at least 25 min in 5% milk in TBS-T.
- ✓ 26. Wash twice in TBS-T
- ✓ 27. Add primary antibody. For TDP-43, use the Ab #33 in -20 database, 1:2000 dilution in 5% BSA in TBS-T. Incubate o/n at 4C with agitation.

Day 2

- ✓ 28. Remove primary Ab.
- ✓ 29. Wash 3 times in TBS-T.
- ✓ 30. Incubate for 1h at RT with Secondary Ab Alexa-fluor diluted 1:5000 in 5% BSA in TBS-T. For TDP-43 I have been using anti-rabbit 488 Alexa-fluor.
- ✓ 31. Wash 3 times in TBS-T.
- ✓ 32. Reveal the membrane using the Chemidoc in EtBr room. Use the >Blotting>>Alexa-488 settings.
- ✓ 33. After revealing, save pictures and wash 3 times in TBS-T.
- ✓ 34. Incubate for 4 hours at 4C with TUBB3 antibody (#24 at 4C, dilution 1:2000) in 5% BSA in TBS-T.
- ✓ 35. Wash 3 times in TBS-T.
- ✓ 36. Incubate for 1h at RT with Secondary Ab Alexa-fluor diluted 1:5000 in 5% BSA in TBS-T. For TUBB3 I have been using anti-mouse 647 Alexa-fluor.
- ✓ 37. Reveal the membrane using the Chemidoc in EtBr room. Use the >Blotting>>Alexa-647 settings.